Part E — The Reallocation Memo (35 pts)

Marketing Mix Modeling for GLUCOREN

Write a two-page memo to the GLUCOREN Brand Lead. Readable by a non-technical executive; defensible to a quantitative reviewer

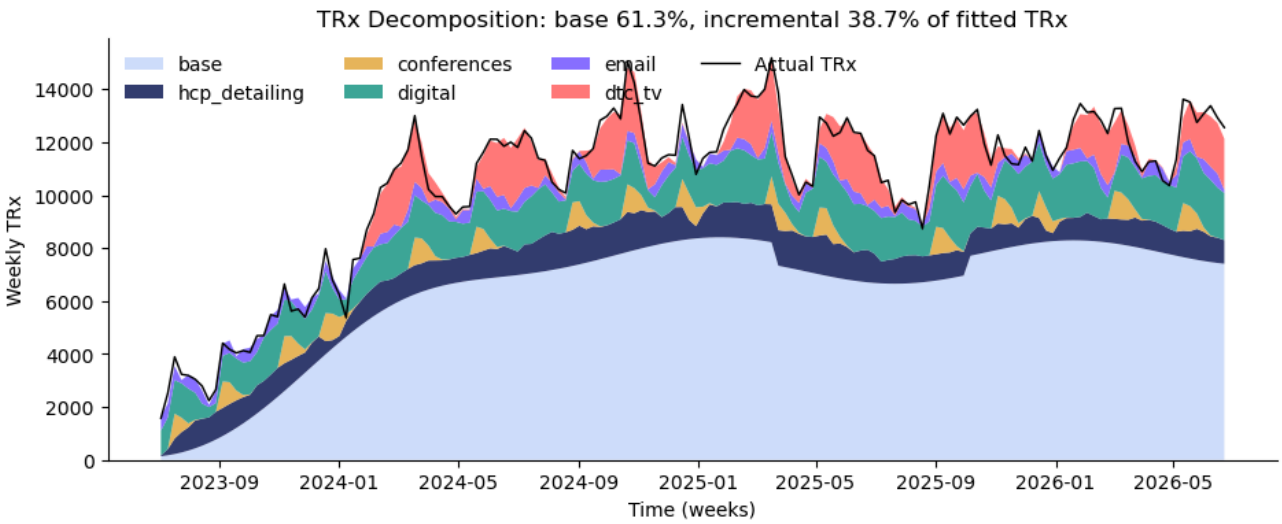
What excellent looks like: Clear TL;DR; reallocation specific, feasible and on a fixed total in two postures; acknowledges what the model misses; ends with a costed experiment.

TLDR:

A one-paragraph TL;DR at the top: what to cut, what to grow, the expected TRx effect with a range, and the one number you are least sure of.

- What to cut: conferences. It ranks last of all our channels spend with a marginal ROI of 0.52, meaning that each dollar spend returns 52 cents. Then after that, cutting off TV spending is second best with a marginal ROI of 1.20 (See Section D10).
- What to grow: email has the highest marginal ROI at 4.04, followed by detailing at 2.79 (See Section D10).
- What the expected TRx effect with a range is (See last code block): Moving $5M (conservative) or $10.3M (aggressive) on a fixed total budget is expected to bring about 39.5k TRx (80% range 32,300 - 46,000, +2.4%) or 59.9k TRx (47,300 - 71,200, +3.7%) over the 3 year period. See code above, which provides a reallocation with specific channels and dollar moves on a fixed total, in two risk postures
    - The constraint that binds first: For the conservative, there is a floor that is being held on conferences ($5.57M); cap is just the $5M budget amount. For the aggresive, there is similarly a floor that is being held on conferences; detailing hits a 1.3x cap, which is flagged in the final section of the code.
    - Note on feasibility: Medical-legal-regulatory review is required for the release of new content in digital/email channels so the timing may limit the rate of spending. Contracts in regards to things like conferences or TV committments may slow us down from acting on our reccomendations. Hiring reps will also take time to train, so budget increases to detailing won't see an immediate effect.
- One number we are least sure of: The detailing ROI. With a high VIF of 19, it is cofounded with ramp (20.3), pbm_preferred (22.8), and digital (21.9). In other words, they share a common time trend where our model can't cleanly separate out the effects of each individual factor.

The Decomposition Chart



Summary: Our base is 61.3% of TRx while media accounts for the rest of the 38.7%; from all media channels, digital (14.3%) and detailing (9.9%) are the largest contributors

What the Model Doesn't Capture

- Some factors aren't being actively captured (i.e. copay programs, samples, prices), so its effects may be seen in the base, ramp, and/or channel coeffs. Panel restatements require that past TRx numbers get revised, so model misalignment may occur and must be carefully disambiguated. MLR cycles, as mentioned previously, limits how rapidly spending can grow/vary. Anything with no variation in the data such as formulary access and detailing spend, by their nature, can't be measured if they never change. Another design thig we referenced earlier in the notebook that isn't captures was hold-out limits (See Section C8).

One Concrete Experiment

- A test which would tighten our least certain estimate, being detailing, would start off by randomly selecting, say 30% of territories, that are matched on baseline prescriptions.
  - Then, we stop detailing for about 14 weeks.
  - The cost would come out to $157.5K per week × 30% × 14 weeks => $0.66M and the lost contribution would be $661,500 × 1.55 ≈ $1.03M, or about 4,100 TRx at $250. Thus, this leaves us with a net cost of $1.03M - $0.66M = $0.36M.
  - If the measured detailing ROI is close to 1.55 or higher, I would keep the planned increase in detailing as our decision rule. But if the ROI is much lower (like below 1.1; similar to TV and conferences), I would shift that increase into email and digital insted.